

Python 10

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Content

1. Uncertainties in Python

Uncertainties in Python

Uncertainties Package

The `uncertainties` package allows you to perform calculations with `uncertainties`. It supports calculations with numbers that have uncertainties and automatically propagates those uncertainties through mathematical operations.

Install the package with `pip`:

```
pip install uncertainties
```

Basic Usage

To use the `uncertainties` package, you need to import the `ufloat` function, which creates numbers with uncertainties:

```
1 from uncertainties import ufloat
2
3 x = ufloat(2.0, 0.1)    # 2.0 +/- 0.1
4 y = ufloat(1.0, 0.2)    # 1.0 +/- 0.2
```

Propagation of Uncertainties

When you perform operations with these numbers, the uncertainties are automatically propagated using linear error propagation

This is great, because you don't have to manually calculate the uncertainties of the results anymore. Just use the `nominal_value` and `std_dev` functions to get the result and its uncertainty:

Propagation of Uncertainties

```
1 # ...
2 z = x + y
3 print(z)    # 3.00 +/- 0.223606797749979
4 print(z.nominal_value)    # 3.0
5 print(z.std_dev)    # 0.223606797749979
6 print(z.n)    # 3.0
7 print(z.s)    # 0.223606797749979
```

Mathematical Functions

The `uncertainties` package supports many mathematical functions, including trigonometric, exponential, and logarithmic functions:

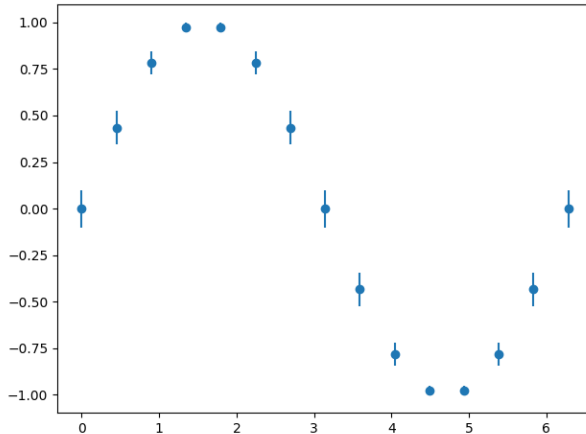
```
1 from uncertainties.umath import sin, cos
2 from uncertainties import ufloat
3
4 angle = ufloat(0.5, 0.05)    # 0.5 +/- 0.05
5 sine_value = sin(angle)
6 cosine_value = cos(angle)
7 print(sine_value)    # 0.48 +/- 0.04
8 print(cosine_value)  # 0.878 +/- 0.024
```


Plotting with Uncertainties

You can also plot data with uncertainties using the matplotlib library:

```
1 import matplotlib.pyplot as plt
2 from uncertainties import unumpy as unp
3 import numpy as np
4
5
6 x = np.linspace(0, 2 * np.pi, 15)
7 xU = unp.uarray(x, 0.1)
8 y = unp.sin(xU)
9 plt.errorbar(x, unp.nominal_values(y), yerr=
    unp.std_devs(y), fmt='o')
10 plt.show()
```

Plotting with Uncertainties



Example: Beer-Lambert Law

The Beer-Lambert law relates the absorption of light to the properties of the material it passes through. It is given by:

$$I(x) = I_0 e^{-\alpha x}$$

where $I(x)$ is the intensity of the light after passing through a material of thickness x , I_0 is the initial intensity, and α is the absorption coefficient.

Example: Beer-Lambert Law

Suppose we have the following measurements:

x	$I(x)$	$\Delta I(x)$
0.0	1.0	0.1
0.5	0.6	0.1
1.0	0.4	0.1
1.5	0.3	0.1
2.0	0.2	0.1

and we want to determine the absorption coefficient α using a fit to the Beer-Lambert law.

Example: Beer-Lambert Law

